

Set 21: Relative atomic mass and mass spectroscopy

Relative atomic mass

Scientists have known for a long time that atoms of different elements had different masses. Originally the lightest element was given a value of 1 and all other elements were measured relative to that. As the mass was a relative value, it had no units. Today, the masses of atoms have been more accurately determined and we know that an atom of hydrogen-1 (H-1) has a mass of 1.67353×10^{-27} kg. These tiny masses are awkward to work with and so scientists still often use relative masses.

The unified atomic mass unit (u), or dalton (Da), is defined as $1/12$ the mass of an atom of C-12 in its ground state and has a mass of $1.6605402 \times 10^{-27}$ kg. Using this value, an atom of H-1 will have a mass of 1.0078 u. (<http://goldbook.iupac.org/U06554.html>)

Mass Spectrometry

Atomic mass spectrometry can be used to qualitatively and quantitatively identify elements present in a compound or mixture. These techniques can be used to measure concentrations as low as a few parts per billion. The basic principle that underlies the use of mass spectroscopy is that when charged particles move through a magnetic field they change direction. When the charge on the particles is the same, particles with the lower mass will experience a greater degree of deflection than heavier particles. When the mass is the same, the particles with a higher charge will experience a greater degree of deflection.

The first step in the process, vaporisation or atomisation, involves the separation of the substance into atoms in the gaseous phase. The atoms are then ionised, usually into +1 ions.

There are a number of ways of ionising a sample. The two methods most commonly used for mass spectroscopy are:

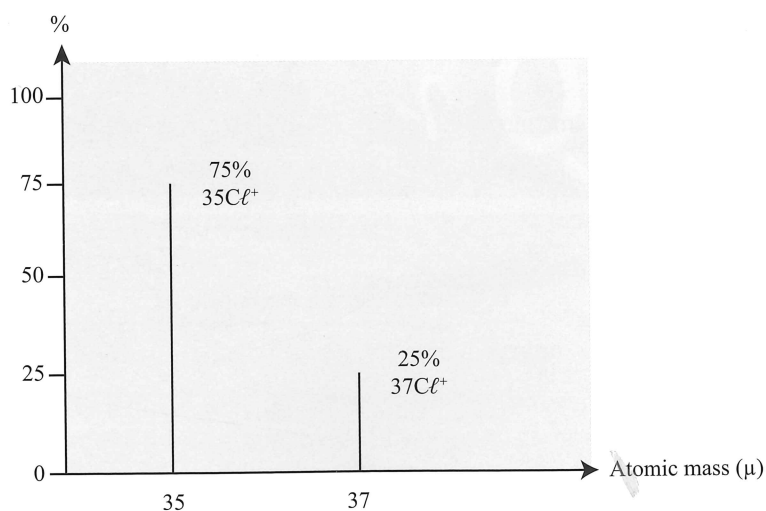
- Inductively coupled plasma (ICP)
- Electric spark

The ions produced will always be positively charged.

Electric sparks were originally used to produce the ions. These have been replaced by ICP. When ICP is used, the atoms are usually converted to singly charged ions.

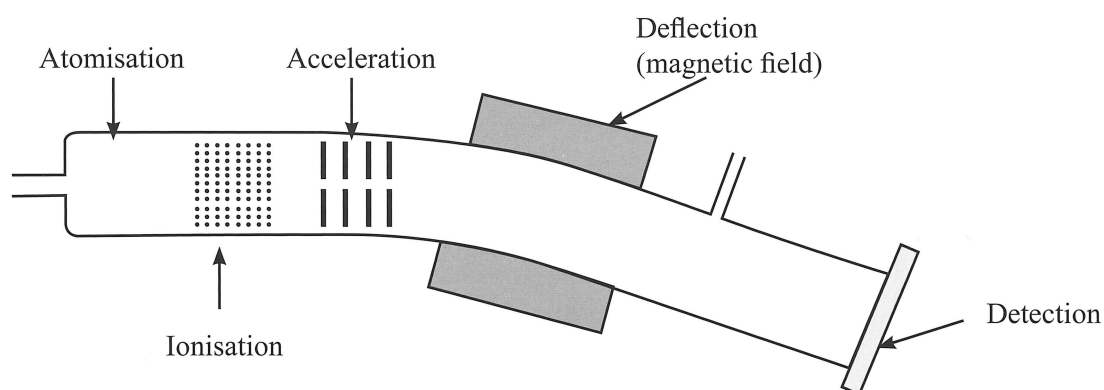
After the sample has been ionised it is passed through a mass analyser. The stream of ions is accelerated through a magnetic field. The charged particles are deflected by the magnetic or electric field. The deflected particles then hit a detector which measures the number of particles and the amount of deflection. From this a mass spectrum is produced.

Mass spectra for Chlorine showing the abundance of Cl-35 and Cl-37.



The separation is based on mass-to-charge ratio, so as long as the charge on the ions is +1, then the mass to charge ratio will be the same as the mass of the ion. For the exercise below, you may assume that the charge on the ions produced is always +1, although you need to be aware that ions with other charges can be produced. The process of acceleration and deflection must occur in a vacuum as there should not be any gas particles that the ions could bounce off.

The diagram below represents the action in a mass spectrometer.



In summary, the process follows the steps atomisation, ionisation, acceleration, deflection and detection. The heavier the ion, the less it will be deflected within the magnetic field. While ions of different charge can be produced in a mass spectrometer, the most common charge is +1.

Calculating relative atomic mass from isotopic composition

Atoms of the same element with a different mass number or number of neutrons are called isotopes. The relative atomic mass and molar mass of elements are derived from the average weight of the various isotopes of an element. Knowing the relative isotopic masses and amounts of each isotope, the relative atomic mass can be calculated.

Atomic mass spectroscopy is an analytical technique that can be used to determine the relative abundances of the isotopes. The technique is described in detail elsewhere.

Examples

Boron consists of two main isotopes, Boron-11 and Boron-10. In a natural sample of Boron, 80.1% will be Boron-11, while the remainder is Boron-10. Calculate the relative atomic mass of boron.

In order to calculate the relative atomic mass, each isotopic mass (which is close enough to the mass number for us to use that value) is multiplied by the percentage and divided by 100.

$$\begin{aligned} A_r(\text{B}) &= \frac{10(19.9) + 11(80.1)}{100} \\ &= 10.8 \mu \end{aligned}$$

Given the relative atomic mass and the isotopes, it is also possible to calculate the relative abundance of them. The simplest examples consist of two isotopes. To determine the relative abundance, simultaneous equations are used. The sum the percentage abundances will always equal 100.

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Boron consists of two isotopes, boron-10 and boron-11. The relative atomic mass is 10.80. What is the percentage abundance of each isotope?

Let x be the percentage of boron-10 and y be the percentage of boron-11.

$$x + y = 100 \quad \text{equation 1}$$

$$10.80 = \frac{10(x) + 11(y)}{100} \quad \text{equation 2}$$

Equation 1 can be rearranged:

$$y = 100 - x \quad \text{equation 3}$$

The value for y can then be substituted into equation 2

$$10.80 = \frac{10(x) + 11(100-x)}{100}$$

The equation can then be arranged to determine the value for x

$$1080 = 10x + 1100 - 11x$$

$$x = 20.0 \%$$

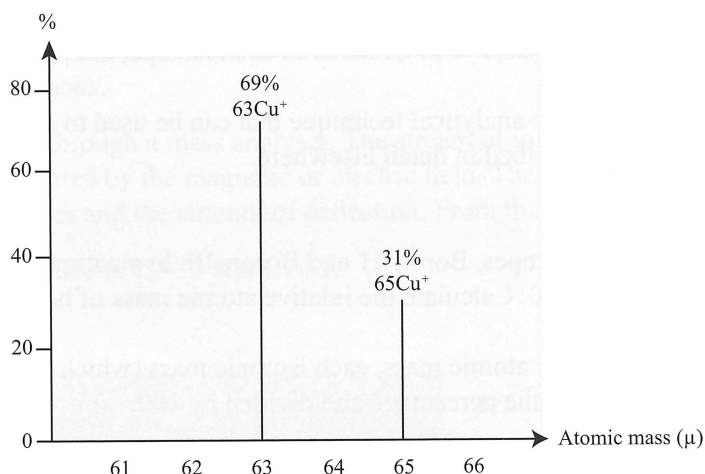
Substitute the value of x into equation 3

$$y = 100 - 20$$

$$= 80.0 \%$$

Atomic mass spectroscopy spectra

A mass spectrum produced will plot the number of ions (relative amount) against the mass-to-charge ratio or mass where the ions are singly charged.



In the example above, a sample of copper was analysed. There are two isotopes of copper present. The peaks at 63 and 65 represent +1 ions. Based on this information, the relative atomic mass of copper can be calculated.

From the spectrum above, the percentage abundance of Cu-63 is 69 and the percentage abundance of Cu-65 is 31. Given this data, we can calculate the relative atomic mass of Cu:

$$\begin{aligned} \text{Ar}(\text{Cu}) &= (69 \times 63) + (31 \times 65)/100 \\ &= 63.62 \mu \end{aligned}$$

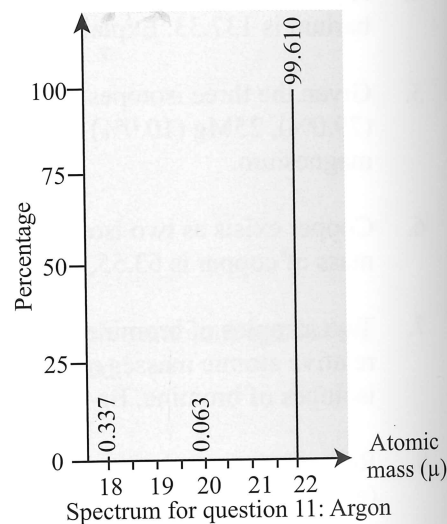
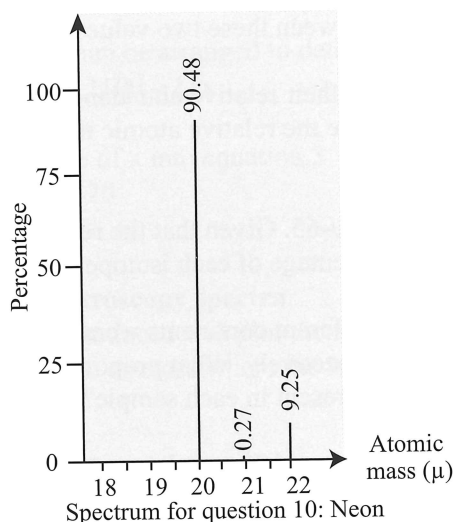
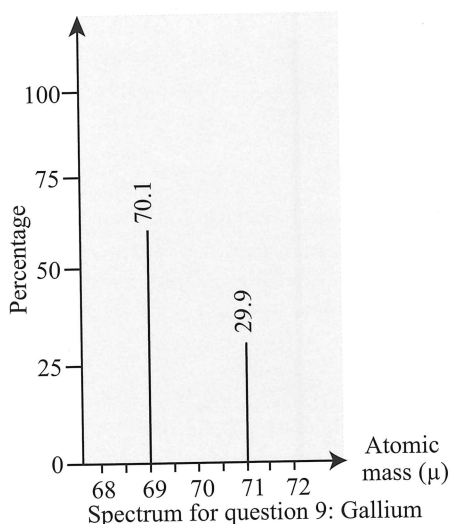
Set 21: Exercises

Notes

1. Chlorine exists naturally as two isotopes, $Cl-35$, which accounts for 75.78% of all chlorine and $Cl-37$, which accounts for 24.22%. Calculate the relative atomic mass of chlorine.
2. There three isotopes of silicon and their relative abundances are: Si-28 (92.2%), Si-29 (4.68%) and Si-30. Calculate the relative atomic mass of silicon.
3. Four isotopes of lead and their relative abundances are Pb-204 (1.40%), Pb-206 (24.1%), Pb-207 (22.1%) and Pb-208 (52.4%). Calculate the relative atomic mass of lead.
4. A particular atom of barium has a mass of 138 while the relative atomic mass of barium is 137.33. Explain the difference between these two values.
5. Given the three isotopes of magnesium and their relative abundances of 24Mg (79.0%), 25Mg (10.0%) and 26Mg, calculate the relative atomic mass of magnesium.
6. Copper exists as two isotopes, Cu-63 and Cu-65. Given that the relative atomic mass of copper is 63.55, determine the percentage of each isotope present.
7. Two samples of bromine, collected from different continents were found to have relative atomic masses of 78.9 and 80.9 respectively. What proportion of the stable isotopes of bromine, Br-79 and Br-81, are present in each sample?
8. Refer to the mass spectroscopy diagram on page 151 to help answer the questions.
 - (a) Briefly describe each of the following processes that occur in a mass spectrometer
 - (i) ionisation
 - (ii) acceleration
 - (iii) deflection
 - (iv) detection
 - (b) The amount of deflection of a charged particle depends on the size of the charge and the mass of the particle.
 - (i) How does the size of the positive charge impact on the degree of deflection?
 - (ii) How does the mass of the particle impact on the degree of deflection?
 - (iii) Which of the following particles will experience the greatest deflection in each case?
 - I $Cu^{+}-63$ or $Cu^{2+}-63$
 - II $B^{+}-10$ or $B^{+}-11$

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9. Gallium has two isotopes. The following mass spectra was produced. Calculate the relative atomic mass of Gallium. (Ga – 69: 70.1 %; Ga – 71: 29.9%)
10. The following mass spectra was produced from a sample of neon. Calculate the relative atomic mass of neon.
20 – 90.48% 21 – 0.27% 22 – 9.25%
11. The following mass spectra was produced from a sample of argon. Calculate the relative atomic mass of argon.
36 – 0.337% 38 – 0.063 % 40 – 99.610 %



Notes